

2.1. Let $(f_n : n \in \mathbb{N})$ be a sequence of measurable functions on a measurable space $(E, \mathcal{E})$. Show that the following functions are also measurable: ⁽¹⁾ $f_1 + f_2$, ⁽²⁾ $f_1 f_2$, ⁽³⁾ $\inf_n f_n$, ⁽⁴⁾ $\sup_n f_n$, ⁽⁵⁾ $\liminf_n f_n$, ⁽⁶⁾ $\limsup_n f_n$. Show also that $\{x \in E : f_n(x) \text{ converges as } n \rightarrow \infty\} \in \mathcal{E}$.

2.1 Note that $[0, \infty]^n$ and $[0, \infty]^\mathbb{N}$ are first countable, so f sequential continuous iff f is continuous

For (1), (2) show that $(a_n, b_n) \rightarrow (a, b) \Leftrightarrow a_n + b_n \rightarrow a + b$ & $a_n b_n \rightarrow ab$

proofs. suppose $(a_n, b_n) \rightarrow (a, b)$, so $a_n \rightarrow a$ & $b_n \rightarrow b$. We know that $a_n + b_n \rightarrow a + b$ &

$a_n b_n \rightarrow ab$, so $E \xrightarrow{(f_1, f_2)} \mathbb{R} \xrightarrow{+} \mathbb{R}$ is measurable $\square$

For (3), (4), (5), (6). We use prop 6.7. by Papadimitrakis [1] which state that for any $f: E \rightarrow \overline{\mathbb{R}}$, f is $\mathcal{E}$ -measurable iff $f^{-1}((a, +\infty]) \in \mathcal{E}$ for all $a \in \mathbb{R}$ where $(E, \mathcal{E})$ is a measurable space.

proofs. Fix $a \in \mathbb{R}$, then $(\sup_n)^{-1}((a, +\infty]) = \bigcup_{n \in \mathbb{N}} f_n^{-1}((a, +\infty])$. Since f_n is $\mathcal{E}$ -measurable $f_n^{-1}((a, +\infty]) \in \mathcal{E}$ and by the countable union closure property, $(\sup_n)^{-1}((a, +\infty]) \in \mathcal{E}$. So $\sup_n$ is measurable.

Notice that $\inf_n (f_n) = -\sup_n (-f_n)$. By (2) and the fact that the constant function $-1: E \rightarrow \overline{\mathbb{R}}$ defined by $x \mapsto -1$ is measurable, we get that $-f_n$ are measurable. By (3), we get that $\inf_n (f_n) = -\sup_n (-f_n)$ is measurable.

$\lim_{n \rightarrow \infty} (\inf_{m \geq n} f_m) = \sup_{n \in \mathbb{N}} (\inf_{m \geq n} f_m)$ & $\lim_{n \rightarrow \infty} (\sup_{m \geq n} f_m) = \inf_{n \in \mathbb{N}} (\sup_{m \geq n} f_m)$ so both are measurable. $\square$

Extra problem. $\max\{f_1, f_2\}$ & $\min\{f_1, f_2\}$ are measurable

proofs. similar to $+$ & $\cdot$ cases, we only need to show that $\max$ and $\min$

are sequentially continuous. we know that $+$, $-$, $| \cdot |$ are continuous and

$$\max\{a_n, b_n\} = \frac{a_n + b_n + |a_n - b_n|}{2} \Rightarrow \max\{a_n, b_n\} \text{ is continuous.}$$

$$\min\{a_n, b_n\} = -\max\{-a_n, -b_n\} \Rightarrow \min\{a_n, b_n\} \text{ is continuous.}$$

$\square$